Active InferAnt Stream 016.2 ~ Generative Education: What Will We Do Wit

ActiveInferAntStream | Generative Education: What Will We Do With Our Faculties? | 1:42:47

All right.
Welcome everyone.
It is December 16th, 2025 and we're live in active inferrant stream 16.2 Two, generative education for today and tomorrow. What will we do with our faculties? This live stream will explore the intersection of human expertise and generative AI in educational content creation, demonstrating how faculty and faculties can leverage cognitive skills and postures and synthetic intelligence tools to create custom modularly configured legibly generated curricula. The presentation will provide open-source resources and methods for people who want to apply these tools today and also hopefully will provide a space for discussion and envisioning the future of generative education and research and what that means for all of us. In the YouTube live chat, please provide any suggestions, questions, comments or prompts. And during the stream we will pick one or more topics to write curriculum for do literature meta analysis on and write a software package and paper. I think we are going to be able to get to all those places. We will look at what is at hand and almost at hand or mandible as it were for three areas of generative education and research and that's going to be curriculum, literature and template. So top level view doxology/curriculum

which is the repo I'm in is going to use
local LLMs to generate custom course
material. So here's the active inference
course which we'll take a deeper look at
and also we can look at generating
introductory biology.
So just drag in an HTML file and here we
have just a four session intro bio
course. So we'll be able to do any
number of sessions, any number of
modules and any types of materials
within the module like lecture lab study
note questions, tests, applications,
extensions, visualizations with mermaid
integrations, further investigations,
open questions at the level of one
module, one session or we could do it
for 50 modules across a 100 sessions or
something like that.
So that is going to be the curriculum
course.
And here's what it looks like to run.
And we'll start this up.
Run pipeline. This is in scripts. Run
pipeline.
And
it will run the test suite. Then it
provides some default courses which are
in the config courses. So we have active
inference 10 modules 20 sessions. Active
inference 15 modules 30 sessions. Intro
bio two modules four sessions. Intro
chem free energy principle 10 modules 20
sessions and intro physics and you can
generate all of them in order or you can
modify it however you want. So let's
let's um await someone's suggestion
and then we will modify one of these.

So we'll come back to that in a second.
The second two repos which I also have open and at the ready are template and literature. So template you might remember from infrarance stream 16.1 about 2 weeks ago. And this is going to be an infrastructure level and a project level for writing a modular manuscript and tested software package that's going to render into all the needed replication unit testing injecting of findings and visualizations into a rendered PDF with citations. So kind of like a power overleaf.
Good things about it. You get to write in markdown. You could do entirely prosbased. You don't need any code or any equations at all. Or if you want to use latte and run a ton of code, you can do that as well. So pretty flexible manuscript rendering. And then split off from that is the repo called literature.
And what literature is going to do after doing the standard testing and everything, you can search for citations through a variety of literature APIs, download the PDFs,
extract their text, and then there's a variety of analyses that are on offer.
And where that goes, we get everything from author lists
to uh embedding analysis. So using LLM to look at where the papers are in latent space to look at the heat map of similarity across a huge numbers of papers
and also look at the principal component analysis different clusters colored by

what year. Look at how different
language use is associated with
different principal components and look
at how different
papers map out differently in that
language space. So supporting rapid
configurable andor faculty in the loop
literature meta analysis. So, where this
is all going and then we'll talk about
what hills we're on and what hills we're
headed for is
kind of a school of tomorrow
infrastructural stack. by no means the
final, by no means the complete stack,
but we have something that with the use
of internet APIs as well as local LLMs
which compress an enormous amount of
background and introductory technical
materials, we can develop curriculum
research and development all around
whatever topic we want. All right. So,
Tucker in the chat has suggested the
history of China economically and tree
grafting from intro to advanced. Let's
work with tree grafting and we will
start with that curriculum. So, we'll
let it chug away in the background while
we explore some other topics. Okay. So,
I'm in curriculum
in scripts run pipeline with Python. And
it's not going to matter which course we
pick because we're going to overwrite
the defaults. So, I'm going to pick one.
The course name
is going to be
tree grafting
from introduction
to advanced

subject expertise area is going to be
trees botany garden
arburitum
science
Course level
graduate
comprehensive
language English. We could do this in
any language. We'll leave that as
default.
Number of modules just so we can kind of
get through it. We'll do five modules.
We'll do four modules with eight
sessions.
Here we can either just hit enter to go
through or you can set a different
number of maximum or minimum topics,
learning objectives, key concepts,
course description. This can be short
and again it provides you the default if
you want to just hit enter and run
through or you can type it in. This
course comprehensively covers everything
you ever wanted to know and a lot you
didn't
know you wanted to know about tree
grafting.
History, biology,
sociology,
esoterica, deep lore, practical,
handson,
agro, economic and cultural.
All aspects
completely covering what the tree gfter
of today and tomorrow
would be best. apt to know today.
Here's additional constraints. This can
be skipped or you can say

course is for
very motivated
tree crafters.
We have access to internet,
all modern technologies and a variety of
kinds of trees and people who need
tree
related services
background
output default and we'll just clear it.
All right. So we are going to in the
first step set up the environment and
run the test that happens before we
configure the course. Then step three is
generate the outline of the course. So
that's what's happening in this first
step. And we can see the logging of the
local lama chugging away. Oh,
where that should go. I guess the course
name
what this we'll we'll we'll let that um
fix. But it generates a outline
and then goes through the outline in
generates primary
materials which are the lecture and the
lab and then generates the secondary
materials
um
uh because okay let's let's
do it a little differently. Let's go in
config.
This will just be a little more
reproducible. Write a new course
on everything needed to know about tree
grafting from basics
through advanced
and then when we run on main it'll be
all properly configured. Um, okay.

Meanwhile, let's start up a meta
analysis for tree graft related
information.
So for this one, I'm going to go to
literature
and I'm going to start a new branch just
called live just so that we don't
mess too much with stuff. So in
literature, we're now on live. So in I'm
going to do run literature.
We are going to clear the library
entirely. Option 7.1
clear. All right, we just have cleared
our library.
We'll delete the extracted text.
Delete the embeddings.
Okay, tree grafting. So, here we go.
Tree grafting fundamentals to advanced
techniques.
Now, we will run the pipeline.
and get that one started.
Okay,
that is going to be
course
7
and we'll just hit enter to go through
all of the defaults.
Okay,
so we've cleared the library
from over here.
and clear the PDFs. Okay, good. All
right. Now, we're going to do meta
analysis 3.1.
We're going to search just on archive
and we're going to turn off the other
sources, but there's a bunch of other
sources. results per keyword. We'll do
50.

Tree grafting.
Okay. There was six papers on archive
for tree grafting. That was step 3.1.
Okay. Move forward to step 4.3.
Now we're going to download the PDFs
from archive
for the tree grafting papers. Those are
going into
the PDFs. Let's see how relevant they
are. Wow. Semantically informed
syntactic machine translation, a tree
grafting approach.
Here's a syntax paper probably based
upon linguistics of grafting.
Nested tree recursion
post league group algebbras
die algebbras
tree
grafting operations
and
Okay, good.
So, scalable hierarchical clustering
with tree grafting.
We will run a meta analysis on those.
And so it's going to analyze
the
papers.
So here we have the different language
use patterns of these six tree grafting
papers.
So
red means overabundance of word use.
Blue is underuse. So PC1
is covering algebra but not tree. PC2 is
negatively loaded on tree, positively on
algebra and so on. Now what's happening
is the Gemma embedding model is being
used to analyze the embeddings of these

six papers. So, kind of quick recap,
when you type in natural language to an
LLM, it tokenizes it, calculates the
embedding, and then generates tokens
from that latent embedding space, pops
back out those tokens to you. So, it
looks like language in, language out.
Embedding is just the first part of that
where we're putting in language in this
case the extracted text of the paper and
getting just the latent space 768
dimensional representation
and then that can be analyzed to look a
little bit more semantically informed
than the PCA. The PCA is looking at word
frequencies and then looking at
orthogonal principal component vectors
that explain or soak up the variance of
the variation in language use at a
syntactic level. Whereas the embedding
space is starting to get us into the
semantics of what the topics are about.
So even if the word usage were non-over
overlapping, for example, different
languages discussing the same topic, we
could still get the embedding profile.
Okay. And
looks like
in our courses,
yep, tree grafting. Here's the tree
grafting outline.
So that is looking good.
Okay. So tree grafting is happening in
the background. We're doing the
embedding analysis. We're almost done
with the embedding analysis on our um
tree grafting algebra. And now meanwhile
I'm going to start up a new branch on

template.
New branch is going to be called tree.
All right.
Now we are going to
go to project within template
manuscript.
So here's the default project
and we're going to bring all this in
comprehensively.
Transform the
project
to be
a variety of useful
methods
and comprehensively
transdisciplinary
theoretical and applied review of tree
grafting.
completely update the
project. So we get wellformed
comprehensive
software package and vast concise
manuscript
regarding all aspects of tree grafting.
We'll use plan mode
that. Okay.
So
now let's return to running the full
pipeline for the tree grafting.
Let it run the test for a few seconds.
Course seven.
Go through the defaults.
Clear.
the outline there.
[snorts] So,
however we'll be most flexible, modular,
robust, general.
Okay,

here what should be the primary focus?
We want
comprehensive review. What should the
package
comprehensive toolkit? Okay.
So now we can look at the embeddings.
We can look at the similarity profiles
of these different papers.
So yellow is the most so on the
diagonal. So we see Baker 2014 and 2015.
They're the most similar which makes
sense. Sounds like it's the first author
is the same. And then we have these
other dare I say trees
dendrograms of the embedding similarity
matrix. And we can also look at the raw
data of that. Here's a graphical
abstract. So this is just sort of
yearbyyear metadata keywords
keywords changing through time author we
only downloaded from archive.
principal components
and the embedding analysis.
So that was the
let's let's do a much larger tree
grafting analysis. So now we're going to
search operation 3.1 and we'll we'll
just allow all of them to be used. Now
many of them are going to reveal papers
that we're not going to be able to get
the full text of. Results per keyword
200
tree grafting.
All right.
We want to
fix all the issues.
So we are going to transform
the template project to be about

grafting. Meanwhile, we have a grafting
related literature metaanalysis
ongoing
and we're fixing a little bit the
curriculum repo so that we can more
easily generate what we already have
generated for the other topics.
So the
plan mode in cursor
is
uh this kind of little little graphical
bug aside is absolutely excellent.
So instead of this kind of treadmill
issue where you could have an LLM going
off the rails and having this kind of
eternal sunshine of the spotless
semantic space or whatever it is,
generating a plan and then repeatedly
returning to the plan is a great
strategy. So we got
curriculum being updated to better
generate as that edge case that we
discovered
transformation of the template repo to
be about grafting only
and we have completed the search phase.
Now we're going to try to download what
we have in our bibliography.
So 4.3 operation. So now there's 453
tree grafting related papers and we're
going to be able to get some of those
PDFs like this first one we got from
Euro PMC. However, there's going to be
other proprietary ones we don't have.
Okay. So let's return to the outline
here and
get through some of these points and
topics. Meanwhile, if anyone has any

other suggestions and we'll periodically
just check back in with all of these
different analyses
and make sure that they're all on the
rails, keep on pushing open source
updates, seeing what people are curious
about. So, here is where we're at. And
to get a better sense of this, let's
just look at the introductory biology
course materials.
So here
um an an important feature also is that
we can modify the outline before
generate. So if we want to look at the
outline of curriculum
output,
we want to go to the biology outline.
So if you look at this and you think I
think I'd rather just do this for module
one. This top level could be written by
hand or it could be modified. So that
first step gives us a a checkpoint and
an opportunity for faculty to to get in
the game and our faculties to get in the
game for looking at the outline of the
course. And then with the primary
materials we generate the um let's try
this again if we can get to the full
tree grafting.
We'll delete it
seven.
Just click through.
All right.
So it's writing the outline. Let's let's
watch that. So um writing the outline is
one place where we can intervene where
we can look at the outline as written
and we can modify the focus or or the

extent or just differently design the
course. So whether it's a one session
educational
presentation on a given topic or you
want the audience to be different, the
language to be different, the points
covered, the constraints, whatever you
want there. All right, there we go. Now
it's
going to push that update since now it's
working in the generalized case
domain. Okay, so now we have this
outline of
The curriculum for
tree grafting
output.
Put that guy away for now.
Tree grafting.
Here's the output.
course outline
tree grafting course outline.
So module one grafting theory basic
grafting whip and tongue
cleft grafting crust bark butdding
bridge
in arching top working framework
troubleshooting and maintenance.
So now that is going to generate for
each of those modules
the primary materials. So it's chugging
away here. We're using the Gemma 3 4
billion parameter language model. So you
can imagine on a bigger better different
computer you could be making these
faster or differently. So first it's
going to go from the outline
and this is happening in
SRC are the modular tested methods and

then scripts have the orchestrator
methods. So three generated outline four
generate primary materials
generates the lecture lab study notes
and
questions
lecture lab study notes diagrams and
questions
for the module. So that's now module one
is complete. Now it's going to move on
to module two and that's going to be
chugging away in the background. And
then after making the primary materials,
it's going to take into the context
window all of the primary materials and
the course outline and then make
secondary materials in the script 5
application, extension, visualization,
integration, investigation, etc. So you
can you can make other kinds of
materials. So that's kind of the
different media that you want to project
the semantics into.
All right. Meanwhile, we have template
still chugging away
on updating the paper
from the example
paper into.
And let's just run that with option nine
just so that it has informiveness
and see how that does.
Okay, it's still writing all the code,
so we won't bug it too much.
Um, okay. So, and then at the end of
generating all the primary and secondary
materials for a course, there's a simple
HTML website, including with dark mode
for just

looking through these materials
and um
using them.
Okay.
So, here's the hill we're on. This is
where we're at. whether in the chat
setting of somebody who's just going to
load up a chat GPT or perplexity window,
whether they're going to drag in a PDF
of a textbook, whether they're going to
use some kind of prompt like you're an
expert introductory biology instructor,
or whether that kind of prompt dispatch
is happening under the hood. So whether
it's that kind of user interface with a
chat or whether it's this more active
niche type generation with the custom
modular orchestration
or whether it's something a little bit
in between with cursor with a rag under
the hood and all that.
We're
already in a situation where you can
make custom generated curriculum for any
topic. And so that's just sort of as if
you were to go to perplexity
and say
make
a complete curriculum for basian
statistics 10 modules.
So
I'm not logged in on this one. So we'll
just see what the free
gives you. So whether you just type in
here's how many modules I want to learn
and here's the topic but however you do
it or whatever services come to exist or
platforms startups government services

etc that offer this we're in a situation
where we can generate any number of
modules we could do 100 modules on
connecting Legos
and it will go into the most
excruciating
fine hairsplitting detail about that. So
any number of modules or sessions
any combination of educational materials
whatever formats we want to project that
semantics into
customizable
and that can be done at the
infrastructural level in a way where the
whole infrastructure could be all
manual. So technically you can have an
all manual generation of course
materials for any topic any group
any language etc or here's the sort of
handsoff version and then this
script kid approach that I'm taking here
gives us intermediate level where we
also get interpretability and these
intermediate artifacts
and so the question is then what and I
think Tucker in the chat sort of comes a
little bit to this which is we make the
curriculum then who does the teaching
ultimately and I think that's that's
just one of the questions is uh how how
is the curriculum
experienced and delivered how how is it
curated what is brought out how is it
evaluated what kinds of domain and
higher order skills are being evaluated
How do we inform and and support
education as a beginning and not not as
just this being the end point hardly

that's the the
open door of the title of what we'll do
with our faculties both the
organizational chart faculty and our
human cognitive faculties and ones that
we also don't know about future
faculties all of those things are coming
into play and we see them chugging away
right in front of us.
Tucker wrote, "Would be neat if
interactive site with homework or
practice online worksheets or
notebooks?" Yes, definitely. Right now
the uh questions part are just static.
But
that can be
intro bio
questions.
Right now it's just static, but it could
be more of an interactive app.
Um,
okay.
So continuing through the outline while
we get template is still chugging away.
So again this just demonstrates you
could look at some of the earlier active
inferrant streams where in the early
days
cursor was a lot more like a musket. It
was like prompt it would make one edit
and then you would assess prompt edit
and then okay now run the test suite.
Now prompt here's the error fix that one
error. But now this is
preparing a plan based upon several
questions.
And this may be using hundreds
or thousands of tools

under the hood.
And that's the importance of starting in
such a heavily scaffolded niche.
dare I say grafting into such a niche
because we already have all the
infrastructure at the top level to
render the manuscript and the style
constraint like hey we use thin
orchestrator scripts to call fully
tested documented methods
which is now
replacing the the previous figures
orchestration script
with
the new methods that are all been
flipped. flipped out with this whole SRC
folder with all this graph data. So that
that still chugs away. Looks good.
Now we have
this 10 module course. We're 70% done
with the primary materials. So again,
we'll just let them both chug away a
little bit.
see how perplexity did here. So here's
that basian probability
question and it'll say write module one
fully all the lecture lab questions.
So it's a little bit like the chat
version
of what here is happening in the sort of
Linux file system nested folder
structure style by iterating over
folders and doing it all in this
versionable space where we can track it
with git line by line. And that's a
great intermediate stop because it plays
nicely with operating systems and git.
And then you can imagine that and there

already are more advanced ways where you
have databases and that gives a little
bit of a more scalable strategy to
versioning material having metadata with
every single row and cell merging all
these different kinds of things that
would be a little bit of a hassle to do
in the file structure setting. It's
possible to do that in a in a more
databaseoriented way and thankfully
that's also easy to flip over. So just
update this from using file structures
into using this database strategy. I try
to show mainly at the file and the
folder level partially because it's uh
again fun investigatable interpretable
at hand and helps bridge the continuity
between these synthetic systems and
augmented coding and just good old
revealing in the finder and just seeing
okay wait these are just plain texts
that are being called by other programs.
So here's the module one and again this
is already a level of educational
accessibility
that's just pending somebody typing in
the questions or even I want to learn
tree grafting
write a course
I do not know where or how to start.
So that to me is the sort of chat
version
of what is possible with an inkling and
an agency to learn. Here's the modules.
That was a fresh chat. Maybe it's
tracking cookies and modules and stuff
like that. Maybe not. And here even we
we see a slightly different

kind of modules, but nonetheless. So
whether again it's that chat format or
whether we're in this
augmented
multiodule investigatable curriculum
setting
that's where we're at. Okay. So while
these are still
okay now it's rewriting the manuscript.
So that is going so it wrote the code it
updated the src
folder first
and the scripts. So src are the tested
methods scripts are the orchestrators of
those methods within the project which
is now grafting.
Now
it's going section by section through
the manuscript files
and rewriting them within this
constrained
setting
with the cursor rules reminding it of
how to actually do citations and all of
that
to update the manuscript. And once that
happens,
we should
be able to regenerate the PDF.
So there's the
tests.
We'll have that
update all the
project related
tests and
documentation to pass.
So now we have two kind of cursor sub
aents going at once. Also, this is a

relatively newish version where you can
see
diff different
just another view on the IDE
editor.
Okay.
Okay. Kind of lost the regular version.
There we go. B control B. All right. So,
we have project tests getting fixed,
manuscript getting written,
and we'll be ready to
run it when it fixes. All right. So in
terms of tools
right now I'm using cursor 2.2
let's confirm exactly
2.2.23
on Mac OSx on this machine.
So each of these three repos, the
literature meta analysis, the project
template and the curriculum development,
they all were made over the last and
versioned over the last several weeks
using this sort of stack with cursor
and using Olama for local LLM calls. All
of the the cursor LLM calls are to the
cloud and the local calls are to mostly
the Gemma 4 billion parameter model, but
I tried a few other ones just to
explore. Each of the repos has a readme.
They're all within the extent of what is
reasonable
aiming to be modular configurable
and to have the local LLM calls with
Olama.
Uh we are generating custom curriculum.
Yes, we are now finishing up.
Now we're into the generation of
secondary materials. So we finished the

primary materials. There's processing
all. So that finished primary. Now it
moves into generating secondary. So now
we're getting the extensions.
So that is going to take in all of the
primary materials from module one as
context and then write several
extensions. So now we're generating
secondary materials.
Um we did a live demonstration of the
course outline generation that was
right here. There's the tree grafting
outline. We could have modified that to
to change the focus. Then we can
generate different content for different
educational contexts.
And I I don't even know if balancing is
the right metaphor, but something like
that, blending, merging, fusing,
surmounting
this possibility
to generate and the the
knowhow and know that and wisdom to know
the direct consequences of manual and
dispatched
educational activities. ities and just
where, how, when, why, where, all that
we blend.
And how do we certify and and mark the
epistemic status of different kinds of
materials? And this is not really
covered in the current um packages other
than just that you can track line by
line where everything is coming from.
But how how do you assess and validate
that something has a a primary factual
accuracy?
um a as well as further topics.

Testing still being improved.
Manuscript almost completed.
So, we should get to that manuscript
pretty soon.
Okay. So, for for those watching,
definitely write any comments or
questions or we can pick a different uh
topic. to explore
central question. What will we do with
our faculty? Whether it's active
inference institute faculty or college
of the redwoods or school of tomorrow or
whatever you are faculty of or at.
And of course the double triple play
with our own faculties they need to be
developed within and across generations.
So what will we do with our faculties
today and where and what will our
faculty be tomorrow?
What is the role of human in AI assisted
education? What is the role of AI in
human assisted education? What is the
role of the human in human education?
What is the role of the AI in AI
education?
Where and how do we have different
standards and quality in spirit and in
letter in fact?
And h how do we signal that higher order
ontology
to have interoperability even within a
given educational institution when there
can be such a a tremendous variety of of
formats? And what what are these
different roles that that faculty might
be in?
I'm very excited to see
how the manuscript goes,

but while it's chugging, I'm not going
to
interrupt.
Okay.
A few different topics just that I wrote
down manually.
while I knew that generating things
would be happening.
So,
Taylor series and generalized
coordinates that'll be familiar if
you've taken the textbook group.
Taylor series being a way instead of
doing explicit prediction
out into future time steps you take a a
series sum over higher and higher
polomial orders. So here the underlying
functions a sine wave we have a first
zeroth order approximation just the
value first order slope second order
quadratic and so on. um and the the
tailor series
and the very related generalized
coordinates
higher moments of statistical
distributions
are where we see a harmonic convergence
of multiscale tactics and strategy as a
nested continuum. So that's not a
situation whereas in the discrete
modeling nested literature we might have
like discrete modeling at the level of
century and then 10 steps within that
the level of decade and 10 within that
at the year 365 within that for the the
day and that would be the all discrete
kind of like a piano with finer and
finer keys more and more explicit

rollouts kind of like a chess algorithm
but when we're in the continuous state
space we see this all at oness of
strategy and tactics. So how do we raise
that and build that awareness to
transfer at the meta and at the domain
level that multi-scale
simultaneous
tactical strategy trim tab steady
cybernetic hand as we move forward and
as students move forward in their own
educations and ways.
Um
tests are passing.
Let's run the rendering.
Okay.
Okay.
We will do
um add coverage to get to 90% testing
successfully.
ensure we get all the way on through
rendering the whole manuscript.
We'll let that one plan out first. Okay.
Meanwhile, we are now deep into
So, six doesn't have secondary
materials, but five we do see the
secondary materials. So, that's still
chugging away locally.
So now we're about 3/4 done generating
this course.
Okay.
There is a recursive and quantum
cognitive effect that arises through
learning and applying active inference.
Not saying that that's the only or the
best way to even point at that moon.
However, when we all of a sudden have
these different domains that we can

learn from at the ready and it's sort of
like well they're all situationally
relevant and important. So what to learn
and and how do we pick what to learn?
There's this really interesting and
helpful
nomadic
reality for learning about active
inference, free energy principle,
synthetic intelligence, cognitive
security, all these kinds of topics. It
helps us in our own navigation of this
space.
And that's this sort of recursive layer
that can be approached from analytical,
computational, empirical, philosophical,
phenomenological,
all these different ways of analyzing
it. And then it's like, oh yeah, we
could
project out the train semantic space
into a more philosophy oriented active
inference course. or we could transpose
this course into another language or
perspective
for sure. That's cool and important even
with a classical introbiology course.
And I guess that's my own statistical
bias uh preference
that there's something very interesting
that I that I look for to other people's
kind of ways of working through that
that when it comes to studying sort of
the the nut of the issue, the kernel of
perception and action and different ways
to study perception and action.
different frameworks for comparing
different kinds of inert on

throughactive systems
in this cyberphysical setting where we
have new flavors of inert on
throughactive.
For example, if we have a text file
within a repo,
it
has a causal influence through cursors
rag
even though it's not a program, but
still having a given piece of semantics
through the soft layer of the LLM
comes to play a causal role. comes to be
a difference that makes a difference
even if it's tucked away in a text file
somewhere which is not exactly
simply the Turing vonoyman type
operational difference and so then it
kind of leads one to think about
a post lisp in a way where we're working
in these repositories
that that blur the line in this sort of
Mcsher Goodell Cherbach style
with what is active and passive, what is
reflexive on what. We can have Olam, the
local LLM,
reviewing its own code
and doing different kinds of projections
and and mirroring of its own code.
So that kind of pattern intuition pump
nexus can apply to quote individual
which is to say top down or seeing the
many as one or the high road as well as
a group level collective intelligence
bottom up low road.
And that is right at the heart of you
know basically is the ant colony the
organism or is the ant colony a

superorganism and how do we talk about
these multiscale systems through through
spaces and times different perspectives
of collective systems which all are. And
then how do we deal with that from the
high road and the low road? And how does
active inference help scaffold that
whole process?
And as
alluded to a few minutes ago, debatably
and engagingly these kinds of topics and
studying the the essence of cognition
itself
in the cyberphysical setting
applies and can be applied relates to
etc. analytical axiomatic ways of
thinking to empirical evidential ways of
thinking and then also to the
phenomenological
interiority view from the inside
metaphysics.
So,
we're in this augmented cognitive niche
where even just simplifying down to me
and my body and this repo
as a as a external
niche to the body. We're going from the
body keeps the score to the body and the
GitHub repo keep the score on through
the niche is not just keeping the score.
It's engaging in these extremely
specific. It's not about more or less
advanced or sophisticated as if there
could be just one measure of that. Like
who's to say that this is more
sophisticated
without defining what that means than a
clam? Um, but a lot is going on under

the hood. So now we're all keeping the score. Everything's keeping the score. Everything's in the game. We're in this massively intraactive setting.
And what does that what does that in general in principle help us reduce our uncertainty about? Well, not too much more thankfully than the posture. It's like saying in the the basketball triple threat, pass, dribble, shoot.
You have that posture because there's degrees of freedom and there's variability of settings and context.
it's like
not resolvable or is resolvable that it isn't resolvable which one of those three options is best because there's locations and settings where even in the same position different people would have different suggestions.
So, cognitive science helps bring us into this posture where expertise can be expressed rather than something like a far farmer's almanac of just giving us the information on which educational materials to create. That I don't think will ever be the case because there's this open-ended non-stationerity
of the educational setting and niches, requirements, all of that. But we're getting to being able to assemble the Megatron in the triple threat
at 100,000x speed
and what is decided upon what we actually present for that tree grafting course. How much more manual work do we want to put in beyond the generation of these modules? Th those are resolved as

unresolvable.
So those are those are the current hills
is how we reckon with these available
local LLMdriven
opportunities.
Let's see [snorts] if we can get at
least a preview
of the graft paper.
still failing a few tests.
So, how do we how do we look from those
hills that we're
working on right now that that are
available again whether to the student
or educator andor educator who has
access to a chat session with an LLM
on through those who are delving deeper
into the infrastructure to use these
repos or other analogous repos.
that make uh more of an industrialized
commoditized approach.
Um the next hills have as many ought as
what ought we do what should we do with
these capabilities as they have is
questions. What what is the case? What
would support active, effective,
accurate, aligned generative synthetic
intelligence? These are technical and
objectively subjective questions.
Whereas the is ought gap remains
undefeated and even having the
confidence like yes we think that we can
generate from this configuration with
this language model hands off or with
these expert educators reviewing it
whatever the case may be. We feel
confident that we can make accurate,
complete, compliant introductory biology
curriculum. So great, how do we go from

that is at the n equals 1 or at the n
equals 100,000 into what we ought to do,
how we ought to
proceed with our tree grafting
education.
Okay. So that completed. Now we have we
have the website for tree grafting
education. Let's look a little bit.
So HTML
just drag it on.
So here's the lecture
key topics
lab
worksheets
study notes key concepts
questions
application
citrus disease
extensions Epigenetics, microbiome, 3D
bioprinting
visualization
autogenerated plain text rendered
mermaid.
Whoa.
integration.
So, here's connecting some dots with
different with the other modules
kind of giving a preview and a review
investigations. How how do
investigations
happen?
like introbiology,
those
findings, those claims in the textbook
can be traced back in in many to most or
all situations to primary experiments
done at some prior point andor ones that
can be done right there

in the lab. So connecting the research
and inquiry part
first principles research and inquiry
like the lab section of a bioclass
connecting that to the literature meta
analysis that's why that gets kind of
partitioned over to the literature repo
um
some of there's some slight differences
in mermaid syntax between
rendering it on GitHub versus the
website But um it's it's out there. Um
so that was the
tree grafting.
Let's make a short course
for
active inference and AI. So,
not that it will hit the grand slam on
the first try, but to give a starting
point for for people to even who would
say, "Oh, no way. I would do it totally
totally differently." It's like, "Great.
Well, put what you want to put on the
table or work with the clay that's on
there." So, let's do write another
course. It's going to be called active
inference AI short.
should cover in
three sessions
total. Three modules.
all the basics and intuitive
and important relevant
features of active inference in
relationship to 2025 slash today's
generative AI landscape
topics also like alignment spatial web
multi- aent emerging
applications etc.

All right.
Meanwhile, we will render R.
Okay.
Get us up to 90% plus coverage in tests.
then
that's a little bit of a strict
um cut off
to have 90% test coverage before even
proceeding with the PDF rendering. But
that's just so that it's faster and
clearer when something is failing lest
we go off really far down a certain road
that's that's untested. But that could
be relaxed. Meanwhile, we will um
keep the 90% and just have it add the
tests. All right. Here we will clear
scripts.
Run the pipeline.
All right. This is active inference for
generative AI.
So that's course two.
So I just typed in two.
Now we could modify
the uh here's here's what the config
file looks like. We could modify the
name, subject, description, number of
courses, etc. or just hit enter just to
go all the way through.
All right, while that is chugging away,
let's see if we have gotten to where we
need here.
Okay, 89% very close. Very close.
um
in arching.
Okay. Previous sessions
explored traditional grafting technique
whip and tongue cleft and saddle graft.
Today we're shifting our focus to a

distinctively distinctly different yet
increasingly valuable method
inarchching. In arching derived from the
Japanese term meaning to weave together
represents a grafting technique where a
scion is attached directly to a side
shoot of the rootstock.
Let's see if how well we would do on the
questions. Okay,
questions might have a little bit of
context
rot,
but that's the context engineering
question. making sure that everything's
generated from from the right um place
with the right context in there. Um
okay,
here we have the short course is already
chugging away. Let's look at the
outline.
So
curriculum
output AI short modules up outline.
So
you know how many times over the years
since since each one of us probably got
in the game where's the basic course on
active inference or how does this apply
to generative AI?
Not that this is again cannot be
emphasized enough that is where the
faculty come into play. The faculties
come into play is in either knowing
to prefer not to even look and give a
fresh take on presenting out what what
the person thinks uh
such a course should be.
run to the PDF see if it goes now but

here's just a starting point and then
this outline we could have intervened at
that step to to modify so introduction
to active inference active inference and
generative AI
advanced applications in future
directions. Multi-agent,
shared world models, precision waiting,
grounding language, and action.
So may maybe somebody thinks that's the
right outline given the the context and
and the group that they're teaching in
that moment. Maybe it's totally off
base, but that's what ought to be
taught, not what is being generated
right here.
Oh, 8939. So close.
Um, update to add more tests and relax
the requirements to 80%.
Okay,
so now here's the modules.
Just push an update
in the
So here we are. module one.
There's the lecture
lab.
Okay. So, we're going to manipulate a
visual stimulus to systematically
increase or decrease the perceived
surprise relative to location. Record
changes in the reported part per
perception.
Quantify the relationship between
perceived surprise and corresponding
perceptual adjustment. Develop an
intuitive understanding of how action
reduces variational free energy.
Document the observations to illustrate

a predictive coding loop.
Participants are seated comfortably in
front of a computer screen.
software's launch dot stimulus is
displayed
without any intervention. Participants
are asked to report
subjective perception
of the dot's horizontal position left
right on a scale of -3 to plus three.
Then the software moves the dot by one
degree per second. Participants continue
to report subjective perception of the
horizontal position at 30 second
intervals.
then returned to center conduct
debriefing.
How did the participants reported
perception of the dot's position change
when the dot was moving to the right?
What does that suggest about their
predictive model?
How did the participants reported
perception of a dot change when the dot
was returning?
Okay.
Yeah. Funny. Little bit of mixed up with
some basic biology questions,
but the other questions are
accurate.
Okay.
See if that will pass through now.
Okay.
All right. Now we're getting further.
Now we're failing because we're hitting
some missing figures.
Fix all missing figures and needed
references to fully render the

grafting project and paper.
All right.
Okay. So we have curriculum
using local LLM to generate now the
getting into the secondary materials for
this three module active inference
generative AI course.
We have cursor
fixing up the tree grafting paper which
we should be able to get rendering
pretty soon.
Then
forgotten but not forgotten.
We have the attempt to download.
We'll just 167 of the PDFs. We'll stop
that and we'll just run the literature
analysis.
We'll just do 6.1. So it's going to do
literature meta analysis without any
embeddings cuz Olama is busy with the
other curriculum repo.
So here we go.
So now we have meta analysis 459 papers
that just happened.
Here's publications by year tree
grafting.
Here's how many PDFs we have. We can
look at what what completeness we have
across different metadata.
Keywords grafting trees grafted roottock
growth fruit keyword evolution
AJ tree
maybe Dr. tree
was writing about grafting
different venue ciiaiculture
act
bone grafting so PCA analysis we can see
there's two clusters it looks like two

main clusters
of literature
got green cluster up here got a blue
cluster and then a few down here So we
have basically blue and the green have
similar values or you know roughly
similar on PC1 but the green cluster has
high on PC2 language use.
Here's the 3D principal component. We
can kind of see that there's two
separate clusters here. Purple cluster
and a green one.
So PC1
is just loaded up heavily on grafting.
PC2
is negatively loaded on grafting, but
it's positively loaded on bone and bone
grafting.
PC3 is positively loaded on artery. PC4
is loaded up on tree. So you can see
that a paper that's discussing bone
grafting would be positive on PC 1 and
two artery grafting on one and three.
And we can sort of look at the language
use a little bit more resolution
to look at what kinds of words are
positively or negatively associated with
each of those principal components that
are calculated from the papers.
So the principal component one is
talking about bone
grafting
not about rootstock citrus apple orange
but
PC
etc.
So, here's bone grafting is this vector
pointing up and to the right.

Grafting in general down into the right.
And then here's trees coming off the
left.
Cool. Very cool.
So, I'll just push that
to the live
branch.
old literature,
but that didn't affect the main.
Okay,
back to template.
Um, okay. Let's
look at
the website for the active inference
course
and then see if we get
Okay.
All right,
now we're getting the grafting
simulations are happening.
Let's see if it will render us the PDF.
Great. Okay,
looks like we should see the manuscript
very soon.
So that's again using the within the
project. So the infrastructure top level
never changes but infrastructure methods
are getting called here to run the test
suite, run the figure generation and
render the manuscript. So here's all the
modular sections about the tree
grafting.
And then the output is going to go into
PDF.
This is rendering the markdowns one by
one.
And then it's going to output a combined
PDF.

up.
Okay.
It's still hashing out some unresolved.
We We will stick around to to get to
render this PDF. Um
so showing it can take a little bit more
than an hour
for all of the steps to to really fully
gel.
I think that can be improved uh as well.
It's just showing what's out there
right now.
Then we'll return to the active
inference. Okay, we have missing figure.
Okay.
So, there's that
course.
Okay, cool.
Let's see if this will render the PDF.
Then it does uh uses the local LLM to
review. Let's just see if
Oh, okay. Here we go. So, title needs to
be changed.
So we will do
for the
project we have
give
a
informative
title. Right now it is
update the subtitle
author is
inferant number 016.
Okay.
We'll see if there's any other changes
to make. Then clickable
clickable PDF
software package

grafting anatomy
equations
methods.
These citations might be hallucinated.
We'll just we'll check this one for
example. But that's why there's the
literature repo.
Okay,
that's a real one.
That's a real citation. But with the
with the literature repo running then
you would know that it was real
literature
whatever that's worth. So here we have
methodology
here's more equations related to
grafting.
Okay species compatibility matrix.
So I injected the heat map in
temperature effects, humidity effects,
discussion, cultural conclusion,
future acknowledgements,
detailed
economic parameters,
extended techniques,
supplemental results,
biogenetic analysis.
make any other updates to improve the
I'll just dump in all the manuscript
sections.
So,
kind of pulling back here,
there are three repos that we were
bouncing back and forth in this stream.
There was literature, template, and
curriculum. Literature did the
literature meta analysis. Template is
the one that you can use to make the

software package and render the PDF.
And then curriculum is using the local
LLM to generate the course material
types,
audience, language, all that.
Um, some key themes, generative
education. I mean,
education was always generative, always
synthetic, always collective, always
kill creative. And now that's playing
out at a speed and tempo and mode that
boggles the minds. How do we use AI to
create useful materials and and know
when not to? What does human in the loop
mean? Or as Michael Garfield calls it,
human on the loop. How do we maintain
faculty oversight and expertise? And I'm
not just talking about associate or
tenure track. I mean, how do we use our
attention and our faculties to be part
of this co-creation lest we get
sidelined and tossed aside?
customization where in terms of the
initial setup
to the implementation to the
investigation of intermediate file
types, how do we customize and how do we
have the
foresight and the situational awareness
to know what is relevant where kind of
like a pre-relevance realization
theory of mind theory of relevance
realization quality assurance and the
future of teaching.
future of teaching and learning
few different discussion points then I
will look in the chat to see if anybody
has written any comments or questions

maybe push some final updates and then
call it a stream for now. So how can
faculty and faculties leverage AI tools
without comprising educational quality?
What new skills and workflows do
educators need in the age of generative
AI? How do we balance efficiency gains
with pedagogical depth?
What are the ethical considerations in
AI generated educational content?
How can institutions support faculty in
adopting these tools?
So, let's see where we're going to get
today. We're going to get two.
We got a
uh
450 paper literature meta analysis on
tree grafting.
Let's regenerate the paper.
We got a
course
with 10 modules,
10 sessions on tree grafting, all just
spiraling out of a 4 billion parameter
language model.
So not even using internet search on
that, not even using a larger model,
small language model generating this
course and then in the context of a well
ststructured template repo,
we are getting a nicely formatted
paper
that we will look in just a second.
Okay. And we'll we'll let the LLM run
here. So here it's going to write up uh
some peerreview comments and summaries
of the paper and also translate it into
Russian, Chinese, Hindi.

So from about an hour ago,
Tucker's suggestion that we study tree grafting,
we now have a 55page
paper,
the tree grafting science and practice, biological mechanisms, computational analysis, and decision support framework. A comprehensive synthesis of 4,000 years of agricultural knowledge.
Wow.
Tree grafting represents one of humanity's oldest and most sophisticated agricultural techniques with documented use spanning over 4,000 years across diverse civilizations.
This comprehensive transdiciplinary review synthesizes biological mechanisms, historical development, technical methodologies, agricultural applications, economic impacts and cultural significance of tree grafting
and everything is all
uh nicely formatted in this PDF. So we can click around and look at the uh the whip and tongue and and the point of the triple juxtiposition
with curriculum literature and template
is like you can imagine being in the special operations research team or the infrastructure division or knowledge engineer at the school of tomorrow or whatever your institution is going to be. And you could imagine going back and forth in series or in parallel doing research, expanding that meta analysis to include blogs and social media and and all these other kinds of sources. So

going out there, foraging, epistemic
foraging for what is out there, bringing
it in, curating, representing it,
building awareness, communicating that
using that information and that
grounding that you went out and forged
for plus the semantics of existing
information in your language model to
develop curriculum and research. So we
could bring that bib file from the
literature and we could drop the bib
file and the summaries into template and
use that to constrain which citations or
which topics we explore. And then
whether you do this before or after your
paper is out, we we can make curriculum
around material. So it expands the prism
of science communication and
participation and inclusion when we can
develop
custom targeted starting point
curriculum for any topic.
Super fun.
Okay.
So,
right now it's doing the peer review and
then it will do the translation on the
paper. We'll take a look at that, but I
think people kind of get where that
would go.
All right.
So, that gets us through here and I'm
going to copy out any
points
from chat. Okay.
Thank you, Tucker, for the suggestions.
Alex, good to see you. Thank you. Okay,
I'll copy out some questions from a few

if I recognize the names and icons
appropriately. Very, very skilled
teachers and learners.
All right. So, Buzz Toot
writes,
"How does Daniel see tool adoption
changing where we think and not just
wondering about how we think?"
Well,
as as with everything within this
whimsical who's on first style genre,
which hasn't come up in a in a in a few
streams, I know. um it's just just a
starting point and a work in progress.
But when I get a first take on that, I
go from well zooming out from how do we
think is where, why, and when are we
thinking just like it's like saying well
what's next from the triple threat that
again that depends on where you are in
the basketball court and game season and
all that. So where we think
I think that could be ranging from the
most deflationary spatialized like
people could just pop off a voice memo
while they're on public transit or write
uh on a piece of paper or in their phone
as they're waking up. And that thinking
can be part of an incrementally
expanding
active umbrella of
both breadth and depth.
So where we think
it's going to be everywhere and and
I don't even think everywhere and
nowhere. I think everywhere.
What is thinking?
Will these tools put us further out into

the world or will it allow us to dive
deeper into the epistemological world of
refinement or both? How does that map
versus territory differential now play
out?
Yeah.
So I I to the first part of the question
I
hope it would be the possibility for
both and when what's called for
is to go further out and to
do edge computing
where none have gone before.
Great.
And
if there's a vector where that boots on
the ground in the world throwness
is refinement
amazing.
If there are uh divergence kind of like
a golden compass where one of the arms
is going further out into practice and
in into the world out there and the
other arm of that golden compass is
diving deeper into the world of
abstraction, sophistication, refinement,
interior epistemology, private language.
And
if sort of the um
I'm working with a uh
John Dunn sort of compass metaphor here
where here it's two lovers who are the
arms of that compass and they circle
around each other.
In that metaphor, I'm thinking, well,
maybe those two points are one and the
same.
Maybe one of them stays fixed. Maybe

where somebody is in the world stays and they're able to spin circles and have a epistemological perimeter around them. Or maybe somebody stays grounded on the epistemological pivot and they trace a circumference, a great circle out in the world. or maybe they walk kind of one moving forward than the other, zigzagging all the way. So, I think both of these things could happen, especially if you have the when not to use it or when to use it question. Just the question of when at all. So, it's not like we're locked into this cockpit and we must use this AI. We must take the first response that it comes off the presses. This is like when we can have both eyes opens, then we can do any number of those pivot maneuvers. How does the map versus territory play out? Yeah, that's very interesting. I think part of the the fascinating aspects of cyber physical is
restricting to just within this niche.
Let's look at the peer review for our uh paper.
So, I'm going to update
tree.
So, we're going to go to template.
Ironically, branch grafting
tree
output
llm. Okay, so here's the peer review.
This is going to get to the the map territory question.
Okay, got translations
quality review.
Okay. Some sections could be elaborated,

specific issues,
recommendations.
Here's a Hindi translation,
Russian translation,
Chinese translation.
probably are better ways to do
translations, but that's a situation
when we're playing around with this
where the territory of the repo and and
the map become their own sort of
singularity.
And it's one thing to be doing just
symbolic manipulation within
talking about biology labs and what
could be, but it's going to collapse to
actuality, including the open-endedness
of actuality with what you actually have
at hand. So, are we talking about the
map of the territory within the repo,
within the digital, or are we talking
about real maps, real territories?
Um, okay.
Kirby wrote, "Literature analysis seem
to pull in large amounts of paper
outside the focus of tree grafting,
getting into heart surgery big time and
some algebra. A more loquacious prompt
would probably fix that.
Absolutely. Having well first you could
just curate the list of papers or you
could have some secondary thresholds.
Like it's very common in meta analysis
to say we did a keyword search we got a
thousand papers and then we restricted
to only within the last 20 years and
only of this type and then we had 45
papers and then we did this and we did
full data analysis and we had 26 papers

and now we're going to make this meta
analysis based upon 26 papers whatever
it is. So you could do positive or
negative terms or otherwise manually
curate which papers you throw in there.
A teacher would know how to graft in the
field. I.e. curriculum generation does
not replace skills doing heart surgery.
No substitute for practice makes
perfect. Yeah. Whether it's whether it's
the the horiculture or the artery
grafting which is still tree grafting or
the algebra. It's just a topic that we
always explore these multiple overloaded
namespaces
and uh and uh all the delights of syntax
and semantics.
So I think that is
good for now. We have tree everything
was updated the paper is updated on the
tree branch of template. But I'm going
to flip back over to main
and
curriculum is updated with
everything that happened.
Ending the stream. So,
hope that was
useful or funny or interesting or all
the above for those who enjoy.
Certainly, it's been fun over the last
few days and weeks and months and years
to
explore all these different corners and
and what is now possible
and of course what happens next.
So thank you all. Till next time.